Learning with Local and Global Consistency

Zhou, Bousquet, Lal, Weston & Schölkopf — NIPS 2003

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Why semi-supervised? The consistency assumption

[D/A: own this frame: problem setting, local vs global consistency, the two-moons intuition. Suggested skeleton below — keep, rewrite, or replace.]

- Few labels, plenty of unlabeled data — labeling is the expensive part
- Local consistency: nearby points share labels (k-NN already does this)
- **Global** consistency: points on the same structure (cluster, manifold) share labels, even when far apart in straight-line distance
- Claim to preview: a classifier that is *smooth on the data's own graph* exploits both at once

The algorithm in four steps (Section 2)

1. Affinity: $W_{ij} = \exp\!\left(-\|x_i - x_j\|^2 / 2\sigma^2\right)$, $W_{ii} = 0$
2. Normalize: $S = D^{-1/2} W D^{-1/2}$, $D_{ii} = \sum_j W_{ij}$
3. Iterate: $F(t+1) = \alpha\, S\, F(t) + (1-\alpha)\, Y$, $\alpha \in (0,1)$
4. Classify: $y_i = \arg\max_c F^*_{ic}$

Y clamps the two revealed labels; every unlabeled point starts at zero and accumulates neighbor evidence.

[D/A: A: add your visual walkthrough / Figure 1 crop here]

How labels spread

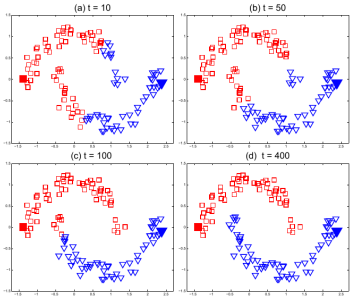


Figure 2: Classification on the pattern of two moons. The convergence process of our iteration algorithm with t increasing from 1 to 400 is shown from (a) to (d). Note that the initial label information are diffused along the moons

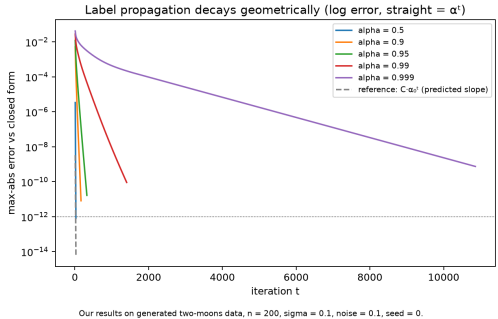
Figure 2: the same iteration at $t = 10, 50, 100, 400$ — Zhou et al. 2003 (paper figure)

- t=10: signal hugs the revealed points
- t=400: labels have flowed *along* the moons, not across the gap
- the graph, not Euclidean distance, decides who is close

[D/A: A: this is your walkthrough frame — bullets are B's strawman, replace freely]

The quiet hero is the normalization

- S is similar to the row-stochastic $D^{-1}W \Rightarrow$ eigenvalues real, all in $[-1, 1]$, top one $= 1$
- $\alpha < 1$ makes αS a **contraction**: converges from *any* start
- Neumann series: $\sum_i (\alpha S)^i = (I - \alpha S)^{-1}$ $\Rightarrow F^* = (1 - \alpha)(I - \alpha S)^{-1} Y$
- That inverse *is* a diffusion kernel: a label reaches every point along every path, weighted α^{path} — **global** consistency, literally
- Same shape as PageRank (damping, teleport); steps 1–2 are also spectral clustering's construction



log error of the iteration per α; straight = geometric decay — our results

Ridge regression wearing graph clothing

$$\mathcal{Q}(F) = \tfrac{1}{2}\sum_{ij} W_{ij}\left\|\frac{F_i}{\sqrt{D_{ii}}} - \frac{F_j}{\sqrt{D_{jj}}}\right\|^2 + \mu\sum_i \|F_i - Y_i\|^2$$

- smoothness energy + fidelity to the known labels, μ is the dial $(\alpha = \frac{1}{1+\mu})$
- $\partial\mathcal{Q}/\partial F = 0$ *is* the fixed point: iterate = solve $(I - \alpha S)F = \beta Y$ = minimize $\mathcal{Q}$
- convex quadratic $\Rightarrow$ unique answer
- our verification: gradient zero to 2×10^{-17}, 18/18 perturbations raise $\mathcal{Q}$, stopping error = tol/$(1-\alpha)$ exactly

α	iters	diff vs solve
0.5	29	8.1e−13
0.9	167	8.2e−12
0.99	1405	9.1e−11
0.999	10860	7.5e−10

α-sweep on generated two-moons data — our results

What the paper's experiments show

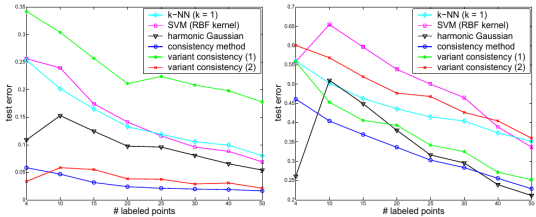


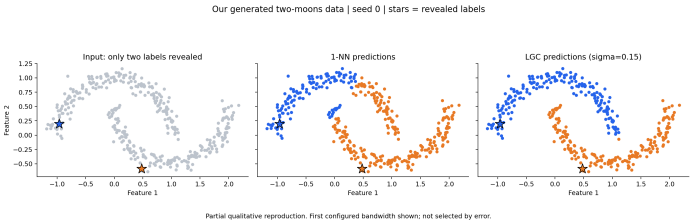
Figure 5: Left panel: the error rates of digit recognition with USPS handwritten 16x16 digits dataset for a total of 3874 (a subset containing digits from 1 to 4). Right panel: the error rates of text classification with 3970 document vectors in a 8014-dimensional space. Samples are chosen so that they contain at least one labeled point for each class.

known to have problems with this method [5]. In our case, jittering by 1 pixel translation leads to an error rate around 0.01 for 30 labeled points.

Figure 5: digits (left), text (right), mean of 100 trials — Zhou et al. 2003 (paper figure)

- $\geq$1 label per class, error on unrevealed points
- digits: LGC clearly beats 1-NN and RBF SVM
- text: the harmonic method wins at some label budgets — no universal dominance
- each method tuned to its own optimum: tuning with few labels stays unresolved

Our paired two-moons experiment



seed 0: 1-NN cuts across the moons; LGC follows the arcs — our results

- 10 seeds, 400 points, **one** revealed label per class
- mean error: 1-NN **22.36%** → LGC **0.45%** (σ=0.15, α=0.99)
- LGC better in every paired trial at 0.15
- σ=0.6: error **32.59%** — bandwidth can bridge the moons

Where this breaks — and what's next

[D/A: own this frame: critique + extensions. Raw material approved in docs/c_paper_evidence.md (limitations 1–8) and results/convergence_check_summary.md (decay-slope nuance, row-scaling equivalence).]

- **[D/A: D: suggest — model selection with almost no labels is still open; our own σ sweep is evidence, not decoration]**
- **[D/A: D: transductive assumption: graph built on the test points themselves]**
- **[D/A: D: dense $n \times n$ solve — cost at scale, what changes for 100k points]**

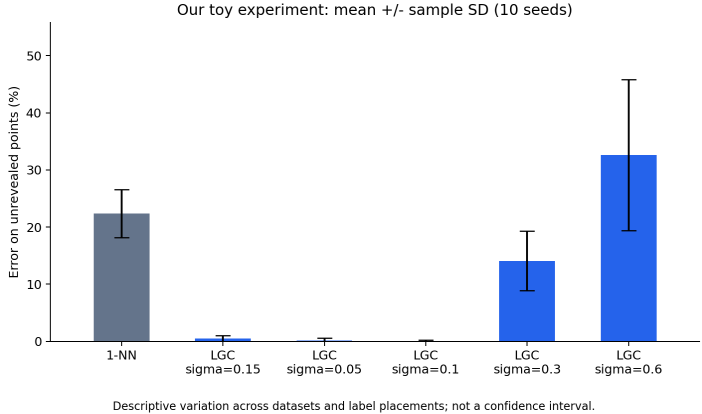
Conclusion

[D/A: D: one slide, four sentences max. Suggested spine: one idea (smooth on the graph = semi-supervised learning), one proof we did ourselves (iterate = solve = minimize, verified to machine precision), one honest limit (bandwidth and budgets), one line on why it still matters (it's PageRank for labels).]

Backup: the equation check (B)

- §2's closing claim: transpose variant "equivalent to $(D - \alpha W)^{-1} DY$" — as matrices it is **not** (measured relative gap 0.82)
- the identity that holds (to 2.5×10^{-15}): $(I - \alpha P^{\top})^{-1} Y = D\,(D - \alpha W)^{-1} Y$
- pre-multiplying by D is a positive *row rescaling* $\Rightarrow$ argmax unchanged — "equivalent for classification" survives, the displayed equation doesn't
- all operator variants shift $\leq 6/200$ toy predictions: classification is robust to normalization choice
- integration receipt: A's core vs B's closed form, $0/400$ label differences, all ten seeds (`results/protocol_check.csv`)

Backup: bandwidth sweep (ours)



mean error ± sample SD over 10 seeds — our results

Sources

- Zhou, Bousquet, Lal, Weston, Schölkopf. *Learning with Local and Global Consistency.* NIPS 2003, 321–328. (Figures 2 and 5: the paper, cited for coursework use)
- Our code, data and figures: `github.com/IliyasBaish/comp3314-group28` (`experiments/`, `results/`, `src/`)
- Ng, Jordan, Weiss (NIPS 2001); Zhu, Ghahramani, Lafferty (ICML 2003) — cited via the paper's Sections 2–4